

Sri Lanka Institute of Information Technology



Sequence Diagram

Web-Based Library Management System

Group ID: MLB-B11G2-10

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Software Engineering | SE2030

B.Sc. (Hons) in Information Technology

Report Overview

This group report follows the Lab Sheet 05 submission structure and is adapted for sequence-diagram documentation. Each member section retains the use-case scenario completed in the previous lab and provides a dedicated reserved space for the corresponding sequence diagram.

Included Content

- Group ID, project topic, and all group members.
- Six completed use-case scenarios, one assigned to each member.
- Six labeled placeholders for the corresponding individual activity diagrams.

UC-01: Manage User Accounts and Roles

Assigned member: Rayyan M.R.M. - IT25101420

Primary actor: Library Administrator

Goal: Create, update, deactivate user accounts, and assign the correct roles.

Preconditions

- The Library Administrator is logged in.
- The Library Administrator has permission to manage user accounts.

Main Flow

1. The Library Administrator selects User Management.
2. The system displays the user-account list.
3. The Library Administrator selects an existing user or chooses to create a new user.
4. The Library Administrator enters or updates user information.
5. The Library Administrator assigns a role: Library Member, Librarian, Finance Officer, or Library Administrator.
6. The Library Administrator saves the changes.
7. The system validates the entered details.
8. The system creates or updates the user account.
9. The system displays a success message.

Alternative Flows

- If required information is missing, the system requests the Library Administrator to complete it.
- If the username or email already exists, the system displays an error.
- If the Library Administrator deactivates an account, the system prevents that user from logging in.

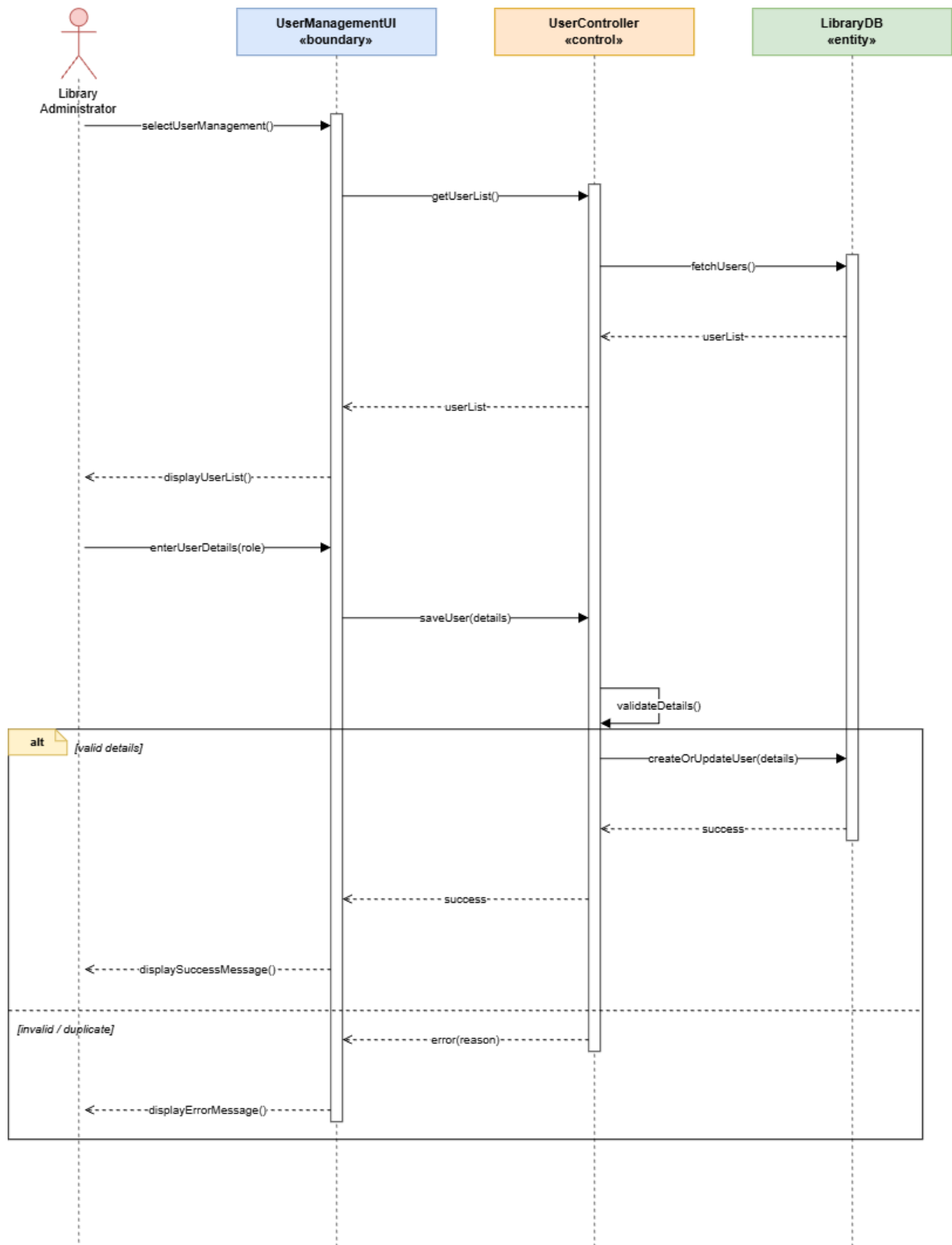
Postconditions

- The user account is created, updated, or deactivated.
- The assigned role determines the user's system access.

Corresponding Activity Diagram

The activity diagram below must model the main flow, decisions, and relevant alternative flows from this use case. Swim lanes are not required for the group-project task.

UC-01: Manage User Accounts and Roles - sequence-diagram



UC-02: Manage Book Catalog and Inventory

Assigned member: Lakshitha P.M.M. - IT25103166

Primary actor: Librarian

Goal: Add, update, search, and remove book records while maintaining availability information.

Preconditions

- The Librarian is logged in.
- The Librarian has permission to manage book records.

Main Flow

1. The Librarian selects Book Catalog and Inventory Management.
2. The system displays the current book catalog.
3. The Librarian selects Add New Book.
4. The Librarian enters the title, author, category, ISBN, number of copies, and other relevant details.
5. The Librarian saves the book record.
6. The system validates the data.
7. The system adds the book to the catalog and sets the availability status.
8. The system displays a confirmation message.

Alternative Flows

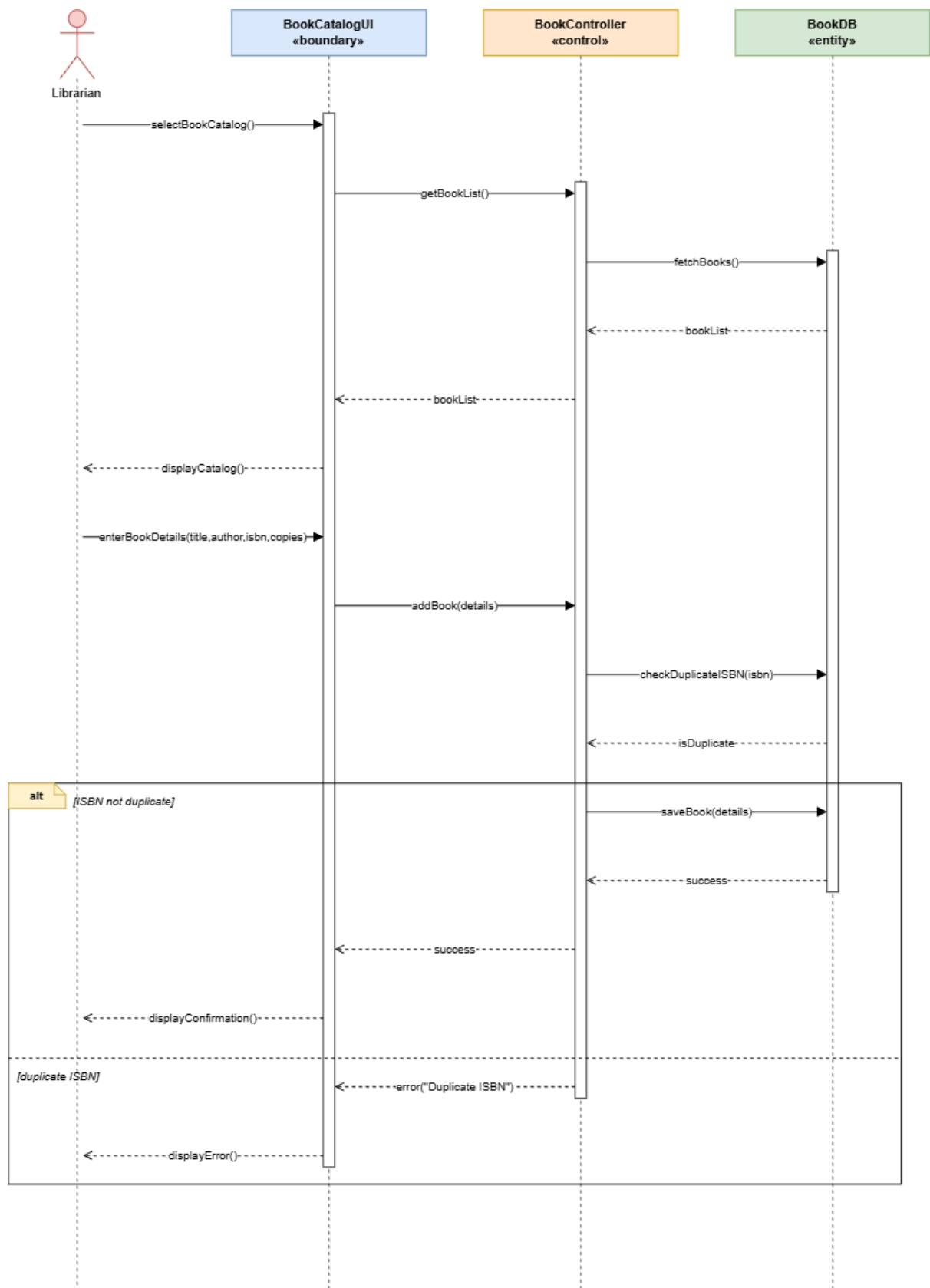
- The Librarian may search for a book by title, author, or category before adding a new record.
- If a duplicate ISBN is entered, the system displays an error.
- The Librarian may update book details or delete a record that has no active borrowing transactions.

Postconditions

- The book catalog and inventory are updated.
- Members and guest visitors can search and view the latest book availability.

Corresponding Activity Diagram

The activity diagram below must model the main flow, decisions, and relevant alternative flows from this use case. Swim lanes are not required for the group-project task.



UC-03: Borrow and Return Books

Assigned member: Wijegunawardana P.K.D.S.U. - IT25102574

Primary actor: Librarian

Supporting actor(s): Library Member; Academic Staff Member (for renewal)

Goal: Issue available books to members and record returned books.

Preconditions

- The Librarian is logged in.
- The Library Member has an active account.
- The selected book is available for borrowing.
- The Library Member has not exceeded the borrowing limit and has no unpaid borrowing restriction.

Main Flow

Borrow Book

1. The Librarian selects Issue Book.
2. The Librarian searches for and selects the Library Member.
3. The system displays the Library Member's borrowing eligibility.
4. The Librarian searches for and selects an available book.
5. The system checks book availability.
6. The system calculates the due date according to library rules.
7. The Librarian confirms the transaction.
8. The system records the borrowing transaction.
9. The system changes the book status to unavailable or reduces the available-copy count.
10. The system displays the due date and confirmation.

Return Book

1. The Librarian selects Return Book.
2. The Librarian enters or searches for the borrowing record.
3. The system displays the borrowed book and due date.
4. The Librarian confirms the return.
5. The system calculates an overdue fine, if applicable.
6. The system updates book availability.
7. The system records the return transaction.
8. The system displays a confirmation and fine amount, if applicable.

Renew Book

1. The Library Member selects Renew Book.
2. The system displays the current borrowed book and due date.
3. The system checks whether another member is waiting for the same book.
4. If no one is waiting, the system extends the due date according to library rules.
5. The system updates the borrowing record.

6. The system displays the new due date and confirmation.

Alternative Flows

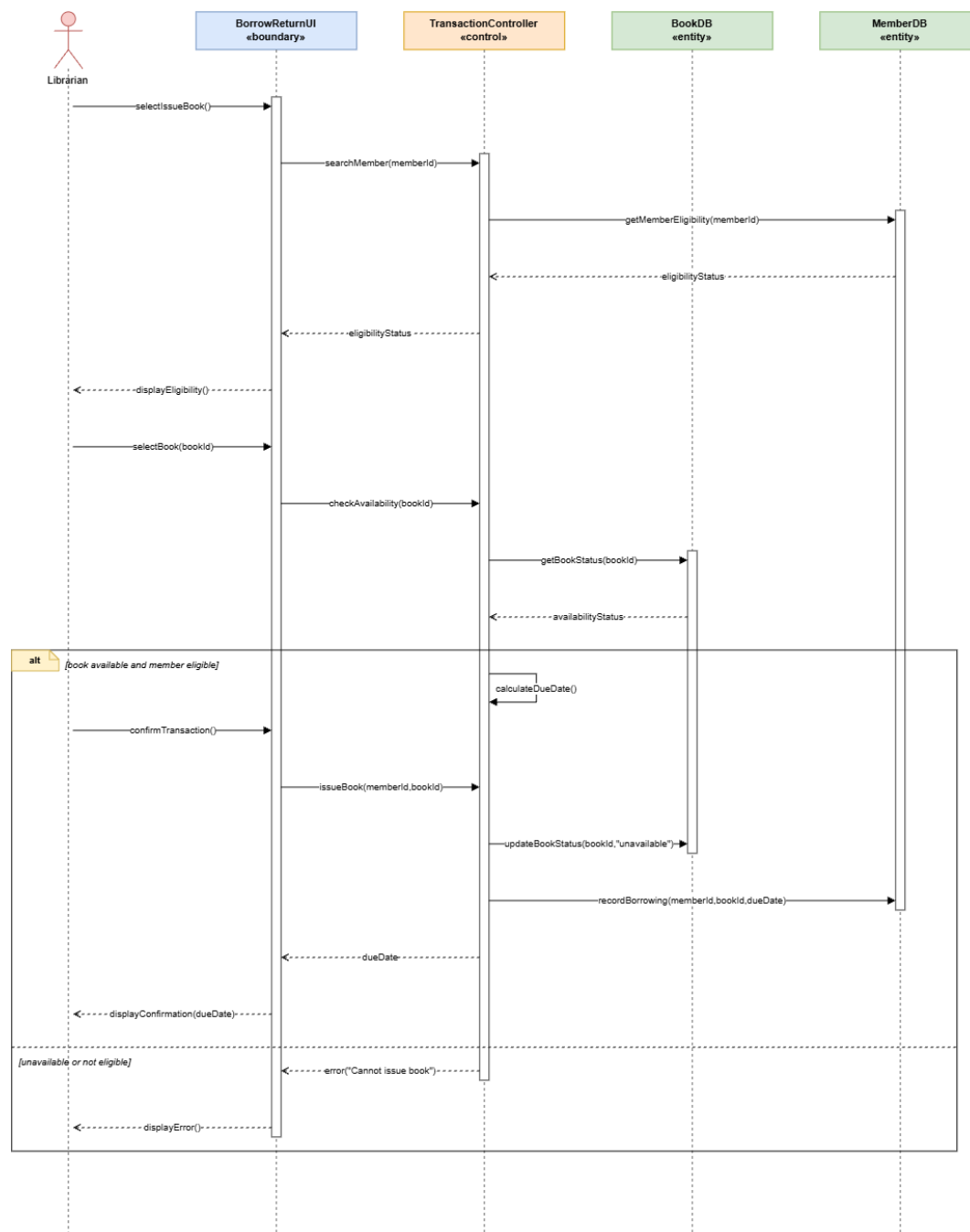
- If the book is unavailable, the system does not allow issuing it.
- If the Library Member has outstanding restrictions, the system prevents borrowing.
- If a fine exists after a return, the system sends the fine record to Fine Management.
- If another member has an active reservation on the book, the system does not allow renewal.

Postconditions

- The borrowing or return transaction is recorded.
- Book availability and the Library Member's borrowing history are updated.

Corresponding Activity Diagram

The activity diagram below must model the main flow, decisions, and relevant alternative flows from this use case. Swim lanes are not required for the group-project task.



UC-04: Manage Book Reservations

Assigned member: Wijewickrama S.N. - IT25100467

Primary actor: Academic Staff Member (a type of Library Member)

Supporting actor(s): Librarian

Goal: Reserve an unavailable book and manage the reservation waiting list fairly.

Preconditions

- The Academic Staff Member is logged in.
- The selected book is currently unavailable.
- The Academic Staff Member has an active account.

Main Flow

1. The Academic Staff Member searches for a book.
2. The system displays that the book is unavailable.
3. The Academic Staff Member selects Reserve Book.
4. The system checks whether the Academic Staff Member already has an active reservation for the same book.
5. The system adds the Academic Staff Member to the reservation waiting list.
6. The system displays the reservation status and queue position.
7. When a copy becomes available, the system notifies the next eligible Academic Staff Member in the waiting list.
8. The Librarian confirms the reserved-book allocation when the Academic Staff Member collects it.

Alternative Flows

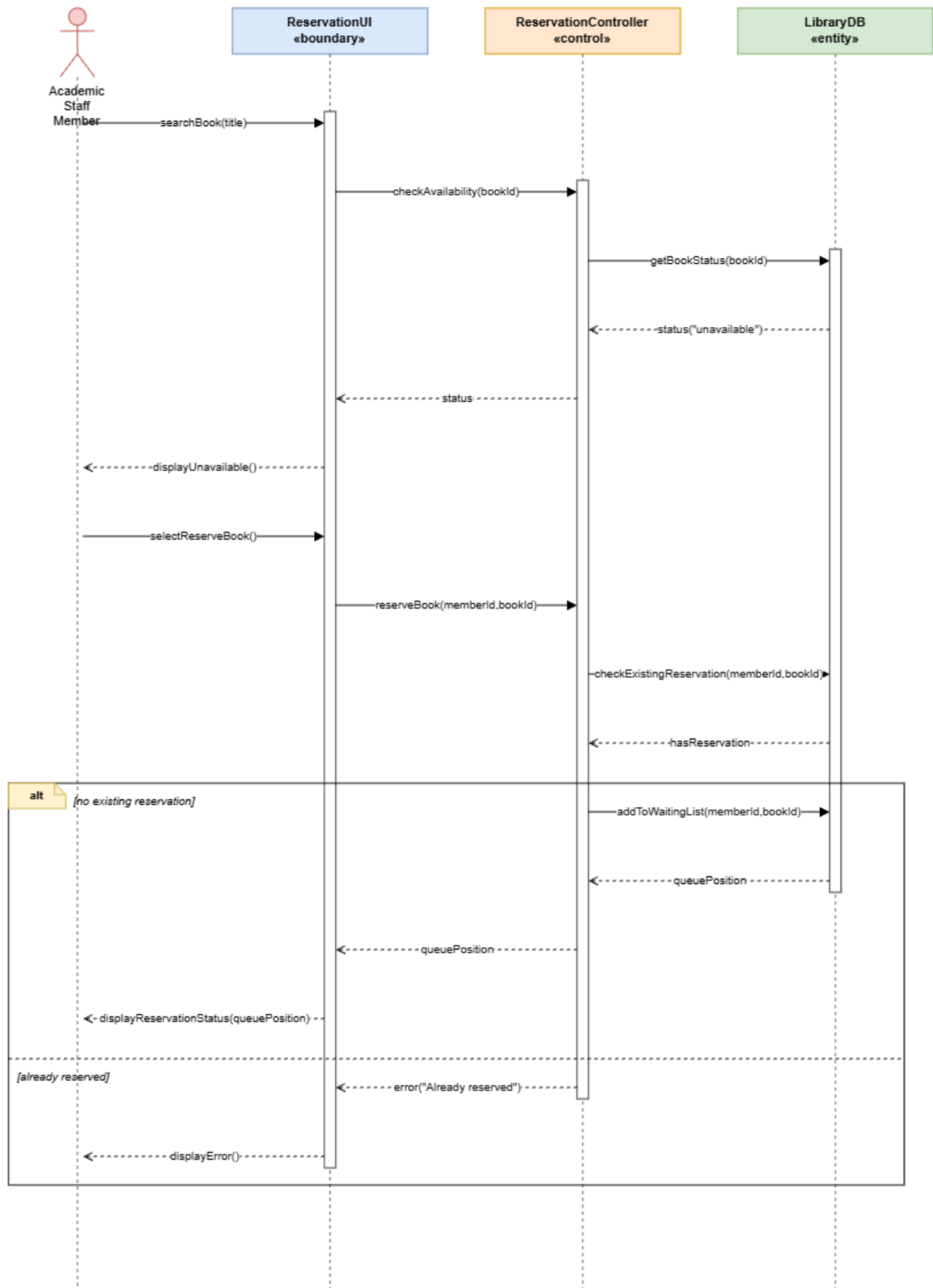
- If the book is available, the system suggests borrowing it instead of reserving it.
- If the Academic Staff Member already reserved the same book, the system displays an error.
- The Academic Staff Member may cancel the reservation before allocation.
- If the Academic Staff Member does not collect the book within the allowed period, the system cancels the reservation and offers it to the next eligible person in the waiting list.

Postconditions

- The reservation status and waiting list are updated.
- The next eligible person receives priority when the book becomes available.

Corresponding Activity Diagram

The activity diagram below must model the main flow, decisions, and relevant alternative flows from this use case. Swim lanes are not required for the group-project task.



UC-05: Manage Fines and Fine Payments

Assigned member: Vithanage C.B. - IT25102212

Primary actor: Finance Officer

Supporting actor(s): Librarian

Goal: View overdue fines, verify automatically calculated fine amounts, record payments, and update payment status.

Preconditions

- The Finance Officer is logged in.
- A Library Member has an overdue-book fine or an existing unpaid fine record.

Main Flow

1. The Finance Officer selects Fine Management.
2. The system displays outstanding fine records.
3. The Finance Officer searches for a Library Member or selects a fine record.
4. The system displays borrowing details, overdue days, calculated fine amount, and payment status.
5. The Finance Officer verifies the fine amount.
6. The Library Member makes the payment.
7. The Finance Officer records payment information.
8. The system updates the payment status to paid or partially paid.
9. The system stores the payment transaction.
10. The system displays a payment confirmation.

Alternative Flows

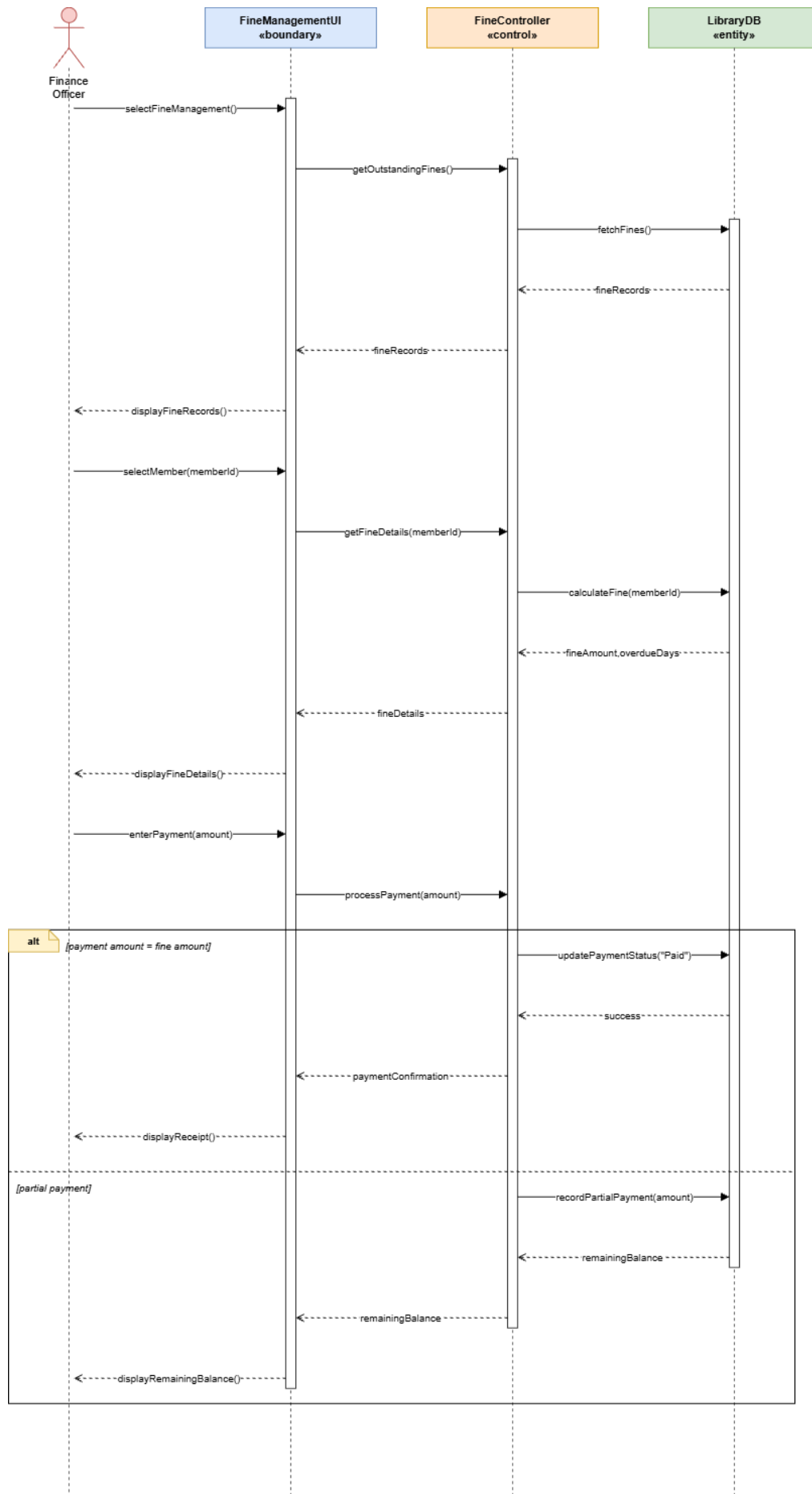
- If no unpaid fine exists, the system displays No outstanding fines.
- If the entered payment amount is less than the fine amount, the system records a partial payment and retains the remaining balance.
- If the calculation is incorrect, the Finance Officer refers the record for correction before accepting payment.

Postconditions

- Fine and payment records are updated.
- The Library Member's outstanding balance is accurate.

Corresponding Activity Diagram

The activity diagram below must model the main flow, decisions, and relevant alternative flows from this use case. Swim lanes are not required for the group-project task.



UC-06: Generate Reports and Analytics

Assigned member: Amukotuwa A.G.S.I. - IT25100041

Primary actor: Library Administrator

Supporting actor(s): Librarian

Goal: Generate reports to support library decisions.

Preconditions

- The user is logged in as a Library Administrator or Librarian.
- The system contains relevant book, borrowing, reservation, user, and fine data.

Main Flow

1. The Library Administrator or Librarian selects Reports and Analytics.
2. The system displays available report types.
3. The user selects a report, such as borrowed books, overdue books, fines, reservations, users, or inventory.
4. The user selects a date range or filtering criteria if required.
5. The system retrieves relevant data.
6. The system generates the selected report.
7. The system displays the report with totals and relevant details.
8. The user downloads or prints the report if needed.

Alternative Flows

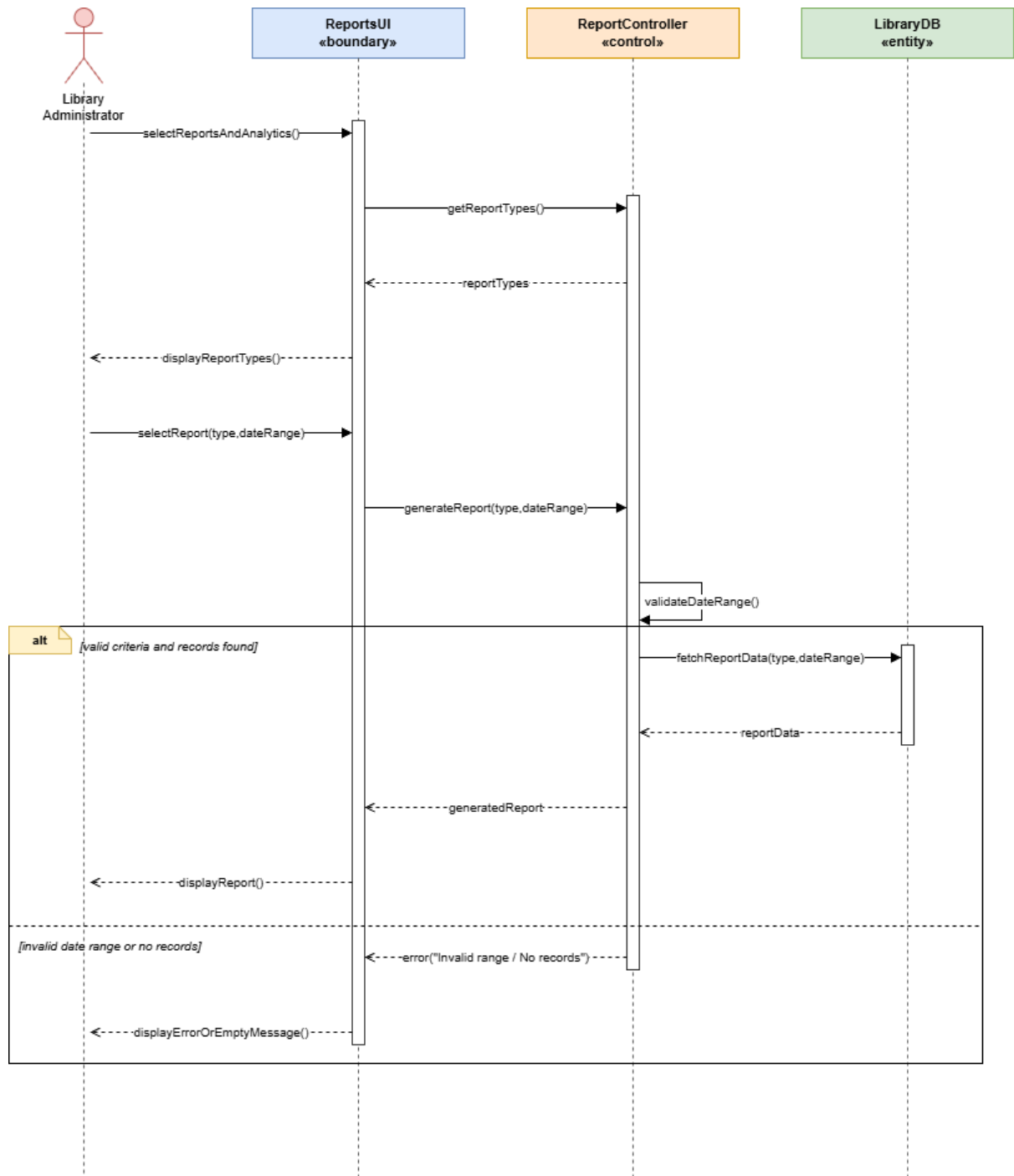
- If no records match the selected criteria, the system displays an empty-report message.
- If an invalid date range is entered, the system requests a valid range.
- The user may change report filters and regenerate the report.

Postconditions

- The requested report is available for viewing, downloading, or printing.
- Management can use the report for operational and financial decisions.

Corresponding Activity Diagram

The activity diagram below must model the main flow, decisions, and relevant alternative flows from this use case. Swim lanes are not required for the group-project task.



UC-07: Manage Book Ratings and Reviewe

Assigned member: Amukotuwa A.G.S.I. - IT25100041

Primary actor: Library Member

Supporting actor(s): Library Administrator

Goal: Allow members to rate and review borrowed books.

Preconditions:

- The Library Member is logged in.
- The Library Member has previously borrowed the book being rated.

Main Flow

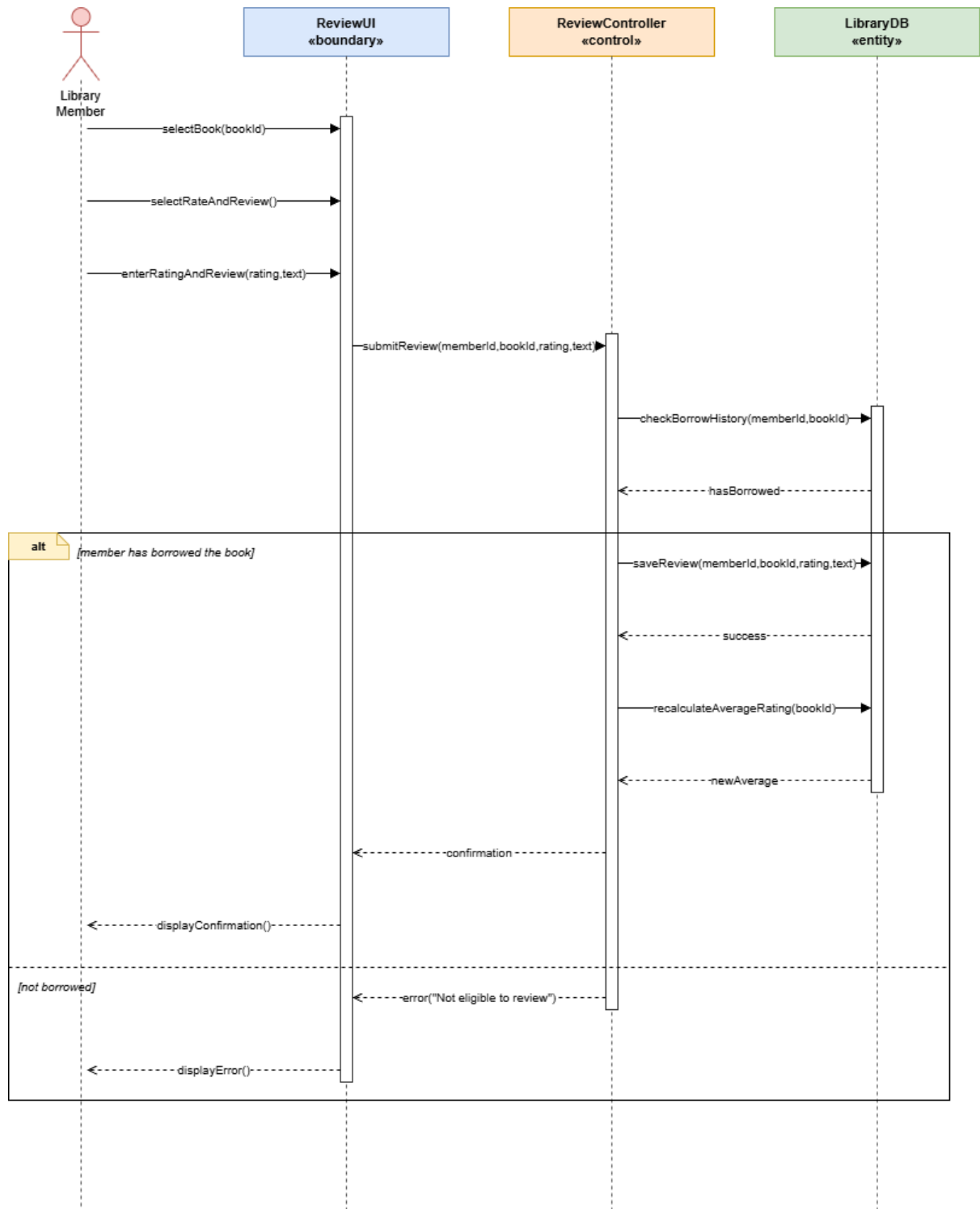
1. The Library Member selects a book from their borrowing history.
2. The Library Member selects "Rate and Review."
3. The Library Member enters a star rating (1-5) and an optional written review.
4. The Library Member submits the review.
5. The system validates that the member has borrowed the book previously.
6. The system saves the rating and review.
7. The system displays a confirmation message.
8. The Library Member or Guest Visitor views a book's details page.
9. The system displays the average rating and list of reviews for that book.
10. The Library Member may view their own review history from their profile.
11. The Library Member selects an existing review from their review history.
12. The Library Member edits the rating or review text.
13. The Library Member saves the changes.
14. The system updates the stored review and recalculates the book's average rating
15. The Library Member selects "Delete" on their own review, OR the Library Administrator selects "Remove" on a flagged review.
16. The system asks for confirmation.
17. The system removes the review and recalculates the book's average rating.

Alternative Flows:

- If the member has not borrowed the book, the system prevents the review submission.
- If the member has already reviewed the book, the system allows editing the existing review instead of creating a duplicate.
- If a review contains inappropriate content, the Library Administrator may remove it.

Postconditions:

1. The book's rating and review data is created, updated, or removed.
2. The book's average rating is kept current for use in reports (e.g., Most Rated Books Report).



UC-08: Manage Member Feedback

Assigned member: Amukotuwa A.G.S.I. - IT25100041

Primary actor: Library Member

Supporting actor(s): Library Administrator

Goal: Allow members to submit general feedback about library services.

Preconditions:

- The Library Member is logged in.

Main Flow

1. The Library Member selects "Submit Feedback."
2. The Library Member selects a feedback category (e.g., Service, Facility, Staff, System).
3. The Library Member enters a comment.
4. The Library Member submits the feedback.
5. The system saves the feedback with status "Pending."
6. The system displays a confirmation message.
7. The Library Administrator selects "Feedback Management."
8. The system displays a list of submitted feedback, filterable by category, date, or status.
9. The Library Member may view the status of their own submitted feedback.
10. The Library Member edits their feedback before it has been reviewed, OR
11. The Library Administrator updates the feedback status (Pending, Reviewed, Resolved) and adds a response.
12. The system saves the changes.
13. The Library Member withdraws their own unreviewed feedback, OR the Library Administrator archives resolved/outdated feedback.
14. The system asks for confirmation.
15. The system removes the feedback record.

Alternative Flows:

- If required fields (category or comment) are missing, the system requests completion.
- If the Library Administrator has already responded, the Library Member can no longer edit the feedback.
- The Library Administrator may filter feedback by status to prioritize unresolved items.

Postconditions:

- Feedback records are created, updated, or removed.
- Feedback data is available for the Feedback Summary Report.

